Week 4 : METADATA

Tools learned:

1. **file**
2. **strings**
3. **exiftool**
4. **hexeditor**
5. **binwalk**

- exiftool

```
┌──(kali㉿kali)-[~/Downloads]
└─$ exiftool Ocean.jpg
ExifTool Version Number         : 13.50
File Name                       : Ocean.jpg
Directory                       : .
File Size                       : 287 kB
File Modification Date/Time     : 2026:04:15 21:55:14-04:00
File Access Date/Time           : 2026:04:15 21:58:35-04:00
File Inode Change Date/Time     : 2026:04:15 21:55:14-04:00
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : inches
X Resolution                    : 115
Y Resolution                    : 115
Exif Byte Order                 : Big-endian (Motorola, MM)
Orientation                     : Horizontal (normal)
Image Width                     : 900
Image Height                    : 557
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Image Size                      : 900×557
Megapixels                      : 0.501

┌──(kali㉿kali)-[~/Downloads]
└─$
```

- can see clearly all info in the image

-hexeditor

```
┌──(kali㉿kali)-[~/Downloads]
└─$ hexeditor computer.jpeg
```

```
File: computer.jpeg                                                    ASCII Offset: 0x00000000 / 0x00002489 (%00)
00000000  FF D8 FF E0  00 10 4A 46  49 46 00 01  01 00 00 01  ......JFIF......
00000010  00 01 00 00  FF DB 00 84  00 09 06 07  13 12 12 15  ................
00000020  12 12 13 15  15 15 17 15  16 16 15 15  18 18 18 17  ................
00000030  18 18 17 16  18 17 17 17  15 17 16 1A  1F 28 20 18  .............( .
00000040  1D 25 1D 15  15 21 31 21  25 29 2B 2E  2E 2E 18 1F  .% ... !1!%)+.....
00000050  33 38 33 2D  37 28 2D 2E  2D 01 0A 0A  0A 0E 0D 0E  383-7(-.-.......
00000060  17 10 10 1A  2D 1D 1F 1D  2D 32 2D 2B  2B 2B 2D 2D  ....- ... -2-+++--
00000070  2D 2D 2D 2B  2D 2D 2B 2D  2D 2D 2D 2B  2D 2D 2D 2B  ---+--+---+---+
00000080  2D 2D 2D 2D  2D 2D 2D 2D  2D 2D 2D 2D  2B 2D 2B 2D  ----------+-+-
00000090  2D 2D 2D 2D  37 2D 2D 37  38 37 FF C0  00 11 08 00  ----7--787......
000000A0  CF 00 F3 03  01 22 00 02  11 01 03 11  01 FF C4 00  ....."..........
000000B0  1C 00 00 00  07 01 01 00  00 00 00 00  00 00 00 00  ................
000000C0  00 00 00 02  03 04 05 06  07 08 01 FF  C4 00 55 10  ..............U.
000000D0  00 01 03 02  02 04 08 07  0B 07 09 08  03 01 00 00  ................
000000E0  01 02 03 11  00 04 12 21  05 31 41 51  06 07 13 22  .......!.1AQ ... "
000000F0  61 71 81 91  32 54 93 A1  B1 C1 D1 14  17 42 52 72  aq..2T.......BRr
00000100  74 92 B2 B3  D2 F0 25 34  62 73 C2 D3  E1 08 15 23  t.....%4bs.....#
00000110  24 35 43 53  82 F1 16 44  64 83 A3 B4  C3 E3 33 94  $5CS ... Dd.....3.
00000120  A2 84 FF C4  00 19 01 01  01 01 01 01  01 00 00 00  ................
00000130  00 00 00 00  00 00 00 00  01 02 03 04  05 FF C4 00  ................
00000140  23 11 01 01  00 02 01 05  01 01 00 03  01 00 00 00  #...............
00000150  00 00 00 01  02 11 03 12  13 21 41 51  31 14 42 71  .........!AQ1.Bq
00000160  F0 22 FF DA  00 0C 03 01  00 02 11 03  11 00 3F 00  ."............?.
00000170  DC 68 50 A1  40 28 50 A1  40 28 50 A1  40 28 50 A1  .hP.@(P.@(P.@(P.
00000180  40 28 50 A1  41 98 70 DF  4D DC A9 F7  C5 BB 61 4D  @(P.A.p.M.....aM
00000190  DA E0 4B 92  E9 4A 8A 96  80 E7 35 09  49 27 25 01  ..K..J....5.I'%.
000001A0  3B F2 AA BE  8F E1 05 EB  CE 25 A4 B1  85 4A 24 27  ;........% ... J$'
000001B0  1B AE 20 12  04 C6 22 DC  4F 45 38 E1  56 88 75 5A  ..  ... ".OE8.V.uZ
000001C0  5E F5 C1 28  68 34 03 8F  49 09 68 AD  86 C0 59 83  ^..(h4..I.h ... Y.
000001D0  26 33 30 33  CA 46 60 55  6B 45 68 25  43 8D 5B B8  &303.F`UkEh%C.[.
000001E0  DA DC 75 45  08 6D 2A 38  C9 6D 60 8B  8C 84 04 94  ..uE.m*8.m`.....
000001F0  E2 21 47 3C  AB 95 CA F9  F2 F6 E1 C7  85 B8 4E 9F  .!G<..........N.
00000200  DF 7F F7 CF  DF F4 DF 38  21 76 5D B3  65 C5 4C 94  .......8!v].e.L.
00000210  99 93 24 10  A2 92 09 DB  11 15 31 55  AE 2E 18 C1  ..$.......1U....
```

-

- binwalk

```
┌──(kali㉿kali)-[~/Downloads]
└─$ binwalk dog.jpg

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------

┌──(kali㉿kali)-[~/Downloads]
└─$ binwalk -e  dog.jpg

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
```

```
┌──(kali㉿kali)-[~/Downloads]
└─$ strings dog_secret.jpg | tail
this is important message
this file will be eliminated
hehe
this is important message
this file will be eliminated
heh
e
```

- binwalk should expose all hidden file in image, so change use of binwalk to strings

- strings

```
┌──(kali㉿kali)-[~/Downloads]
└─$ strings computer.jpeg
JFIF
!1!%)+...
383-7(-.-
-2-+++-----+--+----+---+-----------+-+-----7--787
BRrt
%4bs
#$5CS
!AQ1
5        I'%
&303
F`UkEh%C
8!v]
@W;a
/<@b
C]Yk[\
2%2A
i^Bi
Lb0Z$k
w9?]&?
zU9v
Wm_M
Rz+Z?Z
@PrOeh
NQo^
nrOU
RvjC
n%JQ2I
yA\2
</Ti
]uhV
TFYl
kqcl
K{D'1
E*+(#
5A'Q
=)>l
(Pr_
RkH"1Sd!IP)V
4pq8
eo,Y
YqbuB
][{k
>        >`u
```

- used to extract human-readable text from files

- file

```
┌──(kali㉿kali)-[~/Downloads]
└─$ file solitaire.jpeg
solitaire.jpeg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1×1, segment length 16, baseline, precis
ion 8, 300×168, components 3
```

- file

```
┌──(kali㉿kali)-[~/Downloads]
└─$ file rubiks.jpeg
rubiks.jpeg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1×1, segment length 16, baseline, precision
 8, 233×217, components 3
```